

## FEM Exercise 4

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### 1. Introduction and Geometry Import

The primary objective of this simulation was to evaluate the mixing efficiency of a flow-focusing device integrated with a herringbone mixer (Geometry 3). The CAD files were imported into COMSOL in .dxf format. To define the fluidic channels and isolate the specific device for study, I utilized boolean "Difference" and "Intersection" operations.

To optimize computational resources and ensure a more manageable simulation, the model was split into two distinct components as in figure 1:

- 2D Model: A full geometry used to simulate the initial concentration gradients and baseline flow without the influence of the herringbone structures.
- 3D Segment: A focused model of the herringbone mixer section, used to analyze how the grooves affect fluid mixing.

Two physics interfaces were coupled including **Laminar Flow (spf)** for the velocity field and **Transport of Diluted Species (tds)** for concentration

During the setup, I encountered complications when attempting to crop the 3D segment directly from the original model. To resolve this, I created an additional 2D component by importing and cropping the specific coordinates of the first herringbone section. This cropped rectangle was then extruded and combined with a second import layer to incorporate the 3D herringbone grooves.

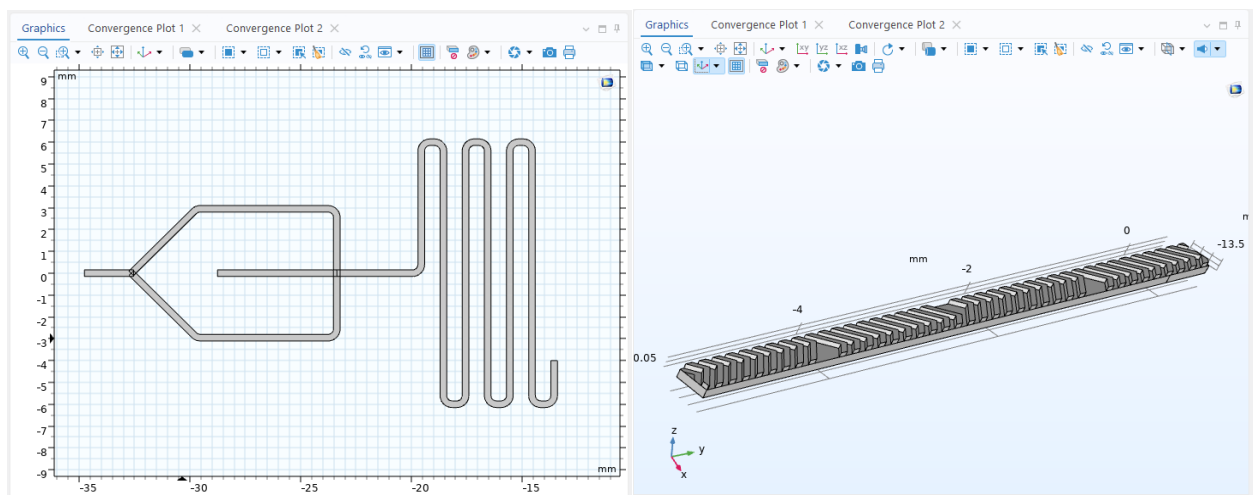


Figure 1: Imported and treated geometry 2D (left) and 3D (right)

## 2. Flow Field Illustration

I have chosen 40 mm/s for the non-diluted species ( $c = 0$ ) channel whereas 10 mm/s for the other ( $c = 1$ ). The concentration when fully mixed is calculated using:  $c_{\infty} = \frac{Q_1 c_1 + Q_2 c_2}{Q_1 + Q_2} = 0.2$ . The flow field was illustrated using streamlines originating from the inlets. The velocity profile shows typical laminar flow.

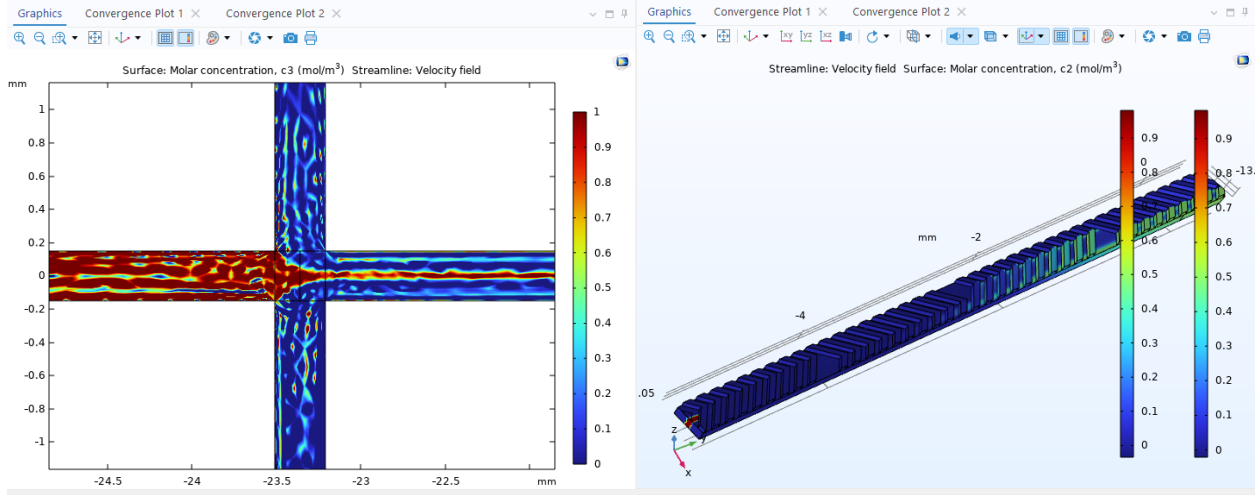


Figure 2: Velocity field illustrated using streamline in COMSOL with 2D and 3D models, respectively.

The velocity fields for both the 2D and 3D models were visualized using streamlines in COMSOL. In the 2D junction, a distinct "needle" of concentration is visible. Because the only factor driving mixing in this 2D environment is molecular diffusion, the mixing speed remains incredibly slow throughout the length of the channel.

This confirms that herringbone grooves are essential for achieving a fully mixed state, as demonstrated in the 3D model. I observed that in the 2D serpentine section, COMSOL tended to assume a fully mixed state by the time the flow reached the 3D input. To ensure the 3D simulation remained a rigorous test of the mixer's geometry, I used the peak concentration recorded at the junction as the input for the 3D model instead. In the resulting 3D simulation, the concentration can be seen stretching and folding toward the end of the segment, indicating effective mass transport.

## 3. Mixing Analysis (Mixing Ratio Calculation)

The mixing ratio  $M$  was calculated using the provided analytical formula:

$$M = \left( 1 - \frac{\int_{outlet} (c - \bar{c}_{\infty})^2}{\int_{inlet} (c - \bar{c}_{\infty})^2} \right) \cdot 100\%$$

Table 1: Comparison between the 2D and 3D models.

| Configuration          | Mixing Ratio (M) | Observation                                                                                                                                                     |
|------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2D channel             | 12.394%          | Mixing is driven only by slow molecular diffusion. The 1D concentration profile at the outlet shows a sharp gradient, indicating poor mixing.                   |
| 3D Herringbone Segment | 89.245%          | Mixing is significantly enhanced. The herringbone structures trigger chaotic advection, folding the fluid layers and increasing the contact area for diffusion. |

The data calculated in COMSOL clearly demonstrates that the inclusion of herringbone grooves leads to a substantial improvement in mixing, jumping from 12% to 89%. This confirms that the grooves successfully induce chaotic advection, which enhances mass transport and creates a much more efficient mixing process.

#### 4. Discussion

The comparative analysis between the 2D baseline and the 3D herringbone mixer reveals a stark performance gap, with mixing efficiency jumping from 12.39% to 89.25%. In the 2D model, the flow is governed strictly by slow molecular diffusion, resulting in a "needle" of concentration with a sharp gradient that persists throughout the channel. Because the flow is laminar and lacks vertical movement, the contact area between fluids remains minimal, leading to the poor mixing observed in the simulation.

The 3D segment overcomes this limitation by utilizing herringbone grooves to induce chaotic advection. These structures physically stretch and fold the fluid layers, exponentially increasing the contact surface area and reducing the distance molecules must travel to diffuse. This mechanism is evidenced by the concentration profile getting "stretched and folded" toward the end of the 3D section, resulting in the high mixing ratio recorded in Table 1.

From a modeling perspective, high Péclet numbers and a refined mesh were critical to ensure the results reflected true convective transport rather than numerical diffusion. Due to computational constraints and software behavior in the 2D serpentine section, a strategic decision was made to use the peak concentration from the junction as the input for the 3D model. This ensured the analysis focused accurately on the mixing support provided by the grooves. Finally, a stationary

solver was utilized to efficiently calculate the velocity field before solving for species transport, providing a robust steady-state analysis.